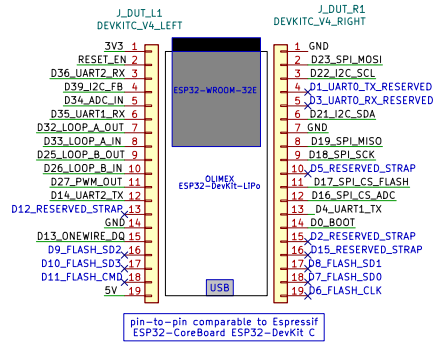
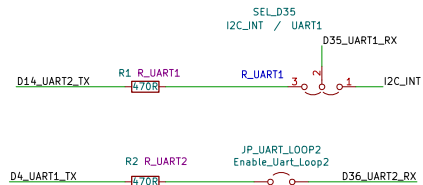


ESP32-DEVKITC V4 DUT SOCKET / HEADER MAP

Two 1x19 sockets represent the DevKitC V4 headers, viewed from the component side with USB at the bottom.
Header labels are the electrical definition:



UART – Full LOOPBACK



RESET / BOOT AUTOMATION PROVISION



FIXED ALLOCATION SUMMARY

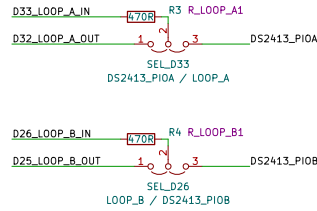
D32->470R->D33 loop A D32->470R->D33 loop A D25->470R->D26 loop B
D27->10k/100nF->D34 + MCP3008 CH0

I2C D21/D22, INT D35 via SEL_D35, FB D39
SPI SCK/MISO/MOSI D18/D19/D23, CS ADC D16, CS flash D17

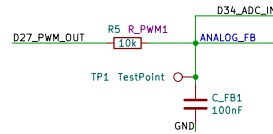
OneWire D13
UART1 TX D4 / RX D35; UART2 TX D14 / RX D36
D35 selects I2C INT or UART1 RX; UART0 D1/D3 reserved for REPL/flashing

No fixed loads on D0/D2/D5/D12/D15.
D6-D11 are module-flash signals and must not be used.

LoopBack Test Selectors – GPIO LOOPBACKS / DS2413 Breakout



PWM / ADC FEEDBACK



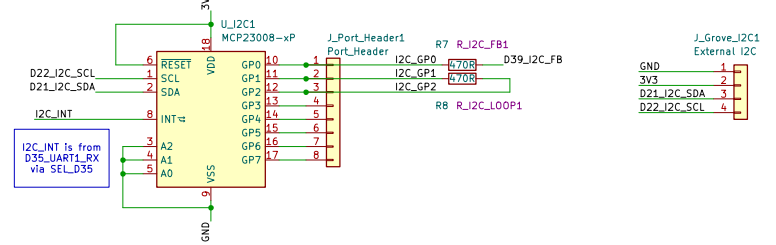
UART2 – External Peer



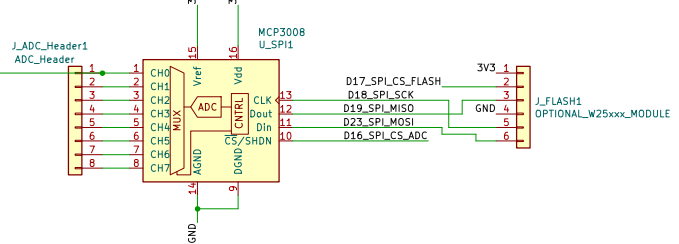
External 5V Supply



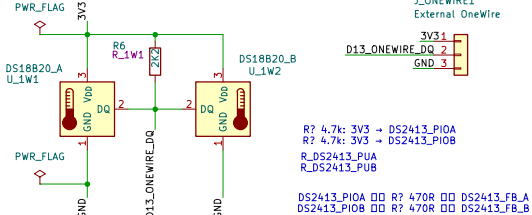
I2C MCP23008 + GROVE I2C Extension



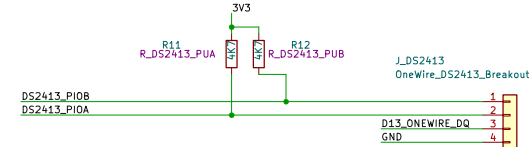
SPI MCP3008 + OPTIONAL FLASH



ONEWIRE – TWO DS18B20 + External Option



ONEWIRE – DS2413 Breakout



I'd retain R3 and R4, relocate them to the common paths, and remove R9 and R10.

See docs/wiring_esp32_devkitc_v4.md
UART0 D1/D3 and boot straps remain reserved
Tests use Espruino Dxx names matching GPIO numbers
ESP-IDF 5 port test harness – fixed GPIO allocation
SimonAndrews

Sheet: /
File: ESP32_V1.kicad_sch
Title: Espruino ESP32 DevKitC V4 Test Harness
Size: A3 Date: 2026-06-21
KiCad E.D.A. 9.0.7

Rev: 1.0
Id: 1/1